

Javokhir Jambulov

Web Engineer

Seoul, South Korea

javokhirjambulov@gmail.com | +82 10-2438-6455 | javokhir.org | [LinkedIn](#) | [GitHub](#)

Profile

Web engineer with 2+ years building Soundcorset's TypeScript frontend and Scala services. Deliver interactive browser tools, offline workflows, authentication, and reliable APIs for a music platform whose Android app has 50M+ downloads.

Experience

Software Engineer | Onsquare | Seoul, South Korea

Aug 2024 – Present

Soundcorset | Web product for sheet music, AI Tutor, recordings, and social features

- **Modernized a legacy frontend** from JavaScript/jQuery to strict TypeScript and Vite, removed CDN dependencies, and organized sheet-music and records code into feature modules.
- **Built sheet-music annotation tools** with Canvas, pressure-sensitive Pointer Events, and offline IndexedDB storage synchronized with AWS S3.
- **Improved multi-page drawing performance** with OffscreenCanvas, Web Workers, lazy initialization, page preloading, and manager disposal; fixed Safari/iPad rendering.
- **Delivered records and social interfaces** for audio playback, AI Tutor evaluations, likes, comments, and follows; centralized playback state in AudioManager.
- **Implemented cross-platform sign-in** with Google and Facebook OAuth across web, Android WebView, and Scala APIs, including token validation and JWT sessions.
- **Built responsive purchase and admin pages** with plan comparison, membership states, and permission-aware controls for read-only and write-capable administrators.
- **Enforced AI Tutor membership access** in Scala/Akka HTTP APIs by returning evaluation charts only to eligible users and driving preview and upgrade states in the UI.
- **Hardened media workflows** across Caffeine, DynamoDB, S3, and SQS by redesigning cache invalidation, using DynamoDB Streams and conditional writes, and preventing async jobs from recreating deleted records; added race-condition and failure-path integration tests.
- **Integrated Google Cloud ModerateText** into the comment lifecycle, implementing allow/hide/block/pending policies, DynamoDB persistence, safe API-failure fallbacks, public-response redaction, admin overrides, and policy/integration tests.
- **Built browser-to-GCP error reporting** with stack traces, build and request context, and duplicate grouping to diagnose production frontend failures.

Technical Skills

Frontend: TypeScript, JavaScript, React, Next.js, Vite, RE:DOM, HTML/CSS, SCSS, Canvas

Browser: OffscreenCanvas, Web Workers, Pointer Events, IndexedDB, WebView/native bridges, responsive UI

Backend & quality: Scala, Akka HTTP, Caffeine, DynamoDB/DynamoDB Streams, AWS S3/SQS, REST APIs, OAuth 2.0, JWT, Google Cloud APIs, ScalaTest

Selected Project

Personal website | Built javokhir.org with Next.js, TypeScript, and Tailwind CSS to present projects and writing.

Education

Bachelor's Degree, Integrated System Engineering | Inha University

2020 – 2024

GPA 4.28/4.5 | Rank 2/35 | Five Academic Excellence Awards

Languages

English (IELTS 7.5) | Korean (TOPIK Level 3)